Nutan Maharashtra Vidya Prasarak Mandal's (NMVPM's)
NUTAN MAHARASHTRA INSTITUTE OF ENGINEERING AND TECHNOLOGY (NMIET)
Under Administrative Support - Pimpri Chinchwad Education Trust (PCET)
ESTD : 1906
Approved by AICTE | Accredited by NAAC ,NBA | Affiliated to SPPU
"Samarth Vidya Sankul", Vishnupuri, Telegaon Dabhade, Taluka Maval, District Pune - 410507
Tel. No. 02114 - 231666 | E-mail : nmiettalegaon@gmail.com | Web : www.nmiet.edu.in

Department of Artificial Intelligence and Data Science

Assignment No.1

Assignment 1 – Fundamentals of Data Science

(Unit I: Fundamentals of Data Science)

Q. No	Question	CO	BT Level
Q1	Explain different types of data: structured, semi-structured, and unstructured with suitable examples.	CO1	BT2 – Understand
Q2	Describe different **scales of measurement** (Nominal, Ordinal, Interval, Ratio) with real-world examples.	CO1	BT2 – Understand
Q3	Explain the **Data Science lifecycle** and compare CRISP-DM and TDSP methodologies.	CO1	BT3 – Apply
Q4	Case Study: A telecom company wants to predict customer churn. Explain the steps of the data science process you would follow to solve this problem.	CO1	BT4 – Analyze

Nutan Maharashtra Vidya Prasarak Mandal's (NMVPM's)
NUTAN MAHARASHTRA INSTITUTE OF ENGINEERING AND TECHNOLOGY (NMIET)
Under Administrative Support - Pimpri Chinchwad Education Trust (PCET)
ESTD : 1906
Approved by AICTE | Accredited by NAAC ,NBA | Affiliated to SPPU
"Samarth Vidya Sankul", Vishnupuri, Telegaon Dabhade, Taluka Maval, District Pune - 410507
Tel. No. 02114 - 231666 | E-mail : nmiettalegaon@gmail.com | Web : www.nmiet.edu.in

Department of Artificial Intelligence and Data Science

Assignment No.2

Assignment 2 – Applications of Mathematical Statistics

(Unit II: Mathematical Statistics in Data Science)

Q. No	Question	CO	BT Level
Q1	Explain vectors and matrices and their importance in data science applications.	CO2	BT2 – Understand
Q2	Solve problems based on **matrix operations** such as addition, multiplication, and determinant calculation.	CO2	BT3 – Apply
Q3	Explain **Bayes' Theorem** and its role in probabilistic modeling with a suitable example.	CO2	BT3 – Apply
Q4	Discuss measures of central tendency (Mean, Median, Mode) and dispersion (Variance, Standard Deviation) and their importance in data analysis.	CO2	BT2 – Understand

Nutan Maharashtra Vidya Prasarak Mandal's (NMVPM's)
NUTAN MAHARASHTRA INSTITUTE OF ENGINEERING AND TECHNOLOGY (NMIET)
Under Administrative Support - Pimpri Chinchwad Education Trust (PCET)
ESTD : 1906
Approved by AICTE | Accredited by NAAC ,NBA | Affiliated to SPPU
"Samarth Vidya Sankul", Vishnupuri, Telegaon Dabhade, Taluka Maval, District Pune - 410507
Tel. No. 02114 - 231666 | E-mail : nmiettalegaon@gmail.com | Web : www.nmiet.edu.in

Department of Artificial Intelligence and Data Science

Assignment No.3

Assignment 3 – Programming for Data Science

(Unit III: Programming for Data Science)

Q. No	Question	CO	BT Level
Q1	Explain the types of machine learning: supervised, unsupervised, and reinforcement learning with examples.	CO3	BT2 – Understand
Q2	Compare **Linear Regression and Logistic Regression** with suitable use cases.	CO3	BT4 – Analyze
Q3	Explain the **Confusion Matrix** and performance metrics such as Accuracy, Precision, Recall, and F1-score.	CO3	BT3 – Apply
Q4	Write Python code using **NumPy and Pandas** to load a dataset and perform basic statistical analysis.	CO3	BT3 – Apply

Nutan Maharashtra Vidya Prasarak Mandal's (NMVPM's)
NUTAN MAHARASHTRA INSTITUTE OF ENGINEERING AND TECHNOLOGY (NMIET)
Under Administrative Support - Pimpri Chinchwad Education Trust (PCET)
ESTD : 1906
Approved by AICTE | Accredited by NAAC ,NBA | Affiliated to SPPU
"Samarth Vidya Sankul", Vishnupuri, Telegaon Dabhade, Taluka Maval, District Pune - 410507
Tel. No. 02114 - 231666 | E-mail : nmiettalegaon@gmail.com | Web : www.nmiet.edu.in

Department of Artificial Intelligence and Data Science

Assignment No.4

Assignment 4 – Data Preprocessing and Visualization

(Unit IV: Data Preprocessing and Visualization)

Q. No	Question	CO	BT Level
Q1	Explain different **data preprocessing techniques** used to handle missing values and outliers.	CO4	BT2 – Understand
Q2	Describe **feature extraction and feature selection** methods with examples.	CO4	BT3 – Apply
Q3	Explain **Exploratory Data Analysis (EDA)** and its role in data understanding.	CO4	BT2 – Understand
Q4	Create visualizations (histogram, scatter plot, box plot) using Python libraries like **Matplotlib or Seaborn** to analyze a dataset.	CO4	BT3 – Apply

Nutan Maharashtra Vidya Prasarak Mandal's (NMVPM's)
NUTAN MAHARASHTRA INSTITUTE OF ENGINEERING AND TECHNOLOGY (NMIET)
Under Administrative Support - Pimpri Chinchwad Education Trust (PCET)
ESTD : 1906
Approved by AICTE | Accredited by NAAC ,NBA | Affiliated to SPPU
"Samarth Vidya Sankul", Vishnupuri, Telegaon Dabhade, Taluka Maval, District Pune - 410507
Tel. No. 02114 - 231666 | E-mail : nmiettalegaon@gmail.com | Web : www.nmiet.edu.in

Department of Artificial Intelligence and Data Science

Assignment No.5

Assignment 5 – Automating AI Workflows with Pandas

(Unit V: Automating AI Workflows with Pandas)

Q. No	Question	CO	BT Level
Q1	Explain the concept of **ETL (Extract, Transform, Load)** and its role in data preprocessing.	CO5	BT2 – Understand
Q2	Discuss **AutoML** and compare it with traditional machine learning approaches.	CO5	BT4 – Analyze
Q3	Explain the role of **open-source tools** such as Python, Jupyter Notebook, and Git in the data science ecosystem.	CO5	BT2 – Understand
Q4	Case Study: A retail company wants to predict future sales using historical data. Explain how regression or classification models can be used for this problem.	CO5	BT4 – Analyze